

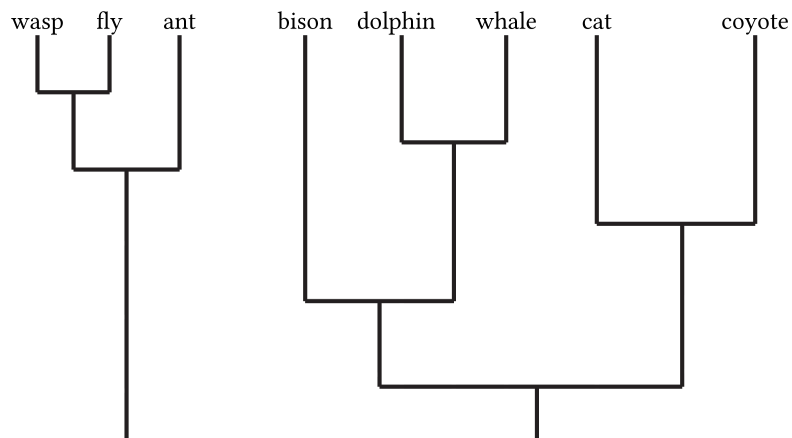
Read each question completely. Explain *thoroughly* your reasoning using clear and concise sentences.

- (10 pts) 1. Draw the entire homology / analogy flow chart that you use to determine whether a trait found in two organisms is a homology or analogy.
- (5 pts) 2. Consider the pharyngeal arches found in embryos of fishes and mammals. During the embryonic stage, the pharyngeal arches do not serve any function. In adult fishes, the pharyngeal arches form the gills. In adult mammals, the pharyngeal arches form a variety of structures, including tonsils, the middle ear, and thyroid glands, none of which are found in fishes. Is this similarity of embryonic pharyngeal arches an example of homology or analogy? *Explain your your complete thought process* as you use the homology / analogy flow chart to arrive at your conclusion.



- (5 pts) 3. Consider the legs of the ant and the cat above. You notice the following structural similarity: *The legs of cats and ants are similar in that they have bendable joints.* Based on this structural similarity, would you conclude the legs similar by homology or analogy? Choose one, and explain your reasoning, using the flowchart as a guide. (5 pts)

Use the following hypothesis to answer questions 4–7 on the next page. *Thoroughly* explain your answers.



- (5 pts) 4. Considering your answer about the ant and cat legs in the previous question, does your conclusion support or falsify the above hypothesis? Explain your reasoning.
- (5 pts) 5. You determine that a bone present in the skulls of the coyote and the cat are homologous structures. Tell whether the homology supports or falsifies the hypothesis. Explain your reasoning.
- (5 pts) 6. Based on this skull homology found in coyote and cat, what *predictions* can you make about the presence or absence of this same homology for the bison, dolphin and whale? Explain your reasoning.
- (5 pts) 7. You discover a second homologous bone in the front legs of the cat and the bison. Based on this homology, what *predictions* would you make about the presence or absence of this homology for the dolphin, whale and coyote, based on the hypothesis? Explain your reasoning.

- (10 pts) 8. Given the following evidence for homology, draw a hypothesis for the following living organisms. Your hypothesis should be in the form of a phylogenetic tree that is consistent with *all* of the data. After each homology, I give the organisms that have that particular homology. If an organism is not listed after a homology, the organism does not have that structure or trait. Do not name any ancestors.

fins: fish, shark

embryonic pharyngeal pouches: fish, shark, frog, mouse, opossum

hair: mouse, opossum

four legs: frog, mouse, opossum